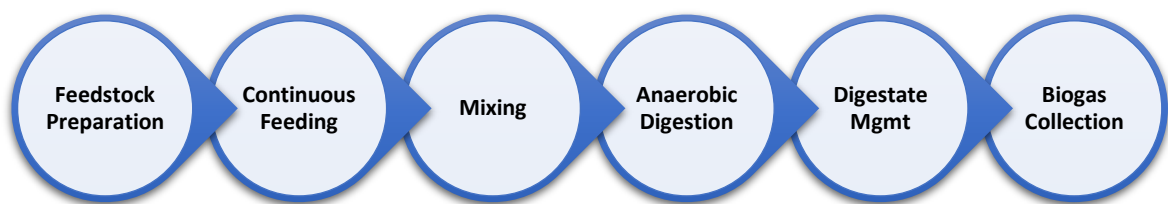


Manufacturing Process and Description of Technology

CSTR Anarobic Digester

The Continuous Stirred-Tank Reactor (CSTR) is a type of bioreactor widely used in CBG plants for the anaerobic digestion of organic waste. In this system, Napier Grass, Cattle Manure and other organic wastes are continuously introduced into a tank where they are mixed with microorganisms that break down the organic matter in the absence of oxygen. Various stages of the CSTR anaerobic digester is presented below.



Some of the key features of the CSTR technology are

- ✓ Continuous Operation
- ✓ Homogeneous Mixing
- ✓ Controlled Environment to maintain optimal pH, Temperature etc.
- ✓ Reduced HRT
- ✓ Scalability

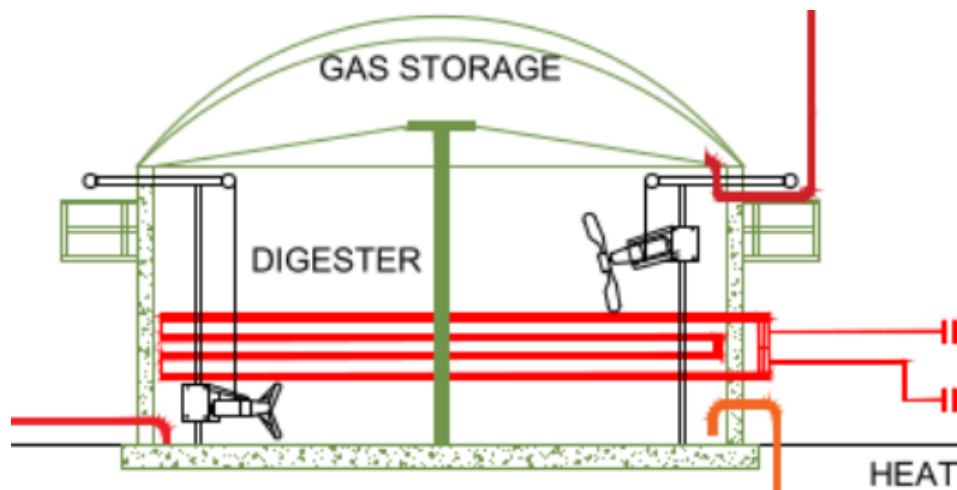


Fig 1.1. A CSTR reactor with mixing, heating, and gas storage provision.

MPSA Two Stage Biogas Upgradation System

Biogas Upgradation plays very important role in achieving the automotive fuel grade quality of product gas known as Bio-Methane / Compressed Biomethane Gas (CBG) / Renewable Natural Gas (RNG). It enriches the quality of the Raw Biogas by removing impurities, including but not limited to moisture, hydrogen sulphide and carbon dioxide. There are various technologies for purification of the Biogas namely Pressure Swing Adsorption (PSA), Membrane Separation, Amine Scrubbing, Water Scrubbing etc.



Fig. 1.2 AtmosPower's Biogas Upgradation System

It is crucial to select the appropriate technology for the CBG project keeping in mind the performance parameter, suitability, reliability, and proven track records. Most of the convention / ordinary / single stage systems are known to have gas losses upto 30%. The biogas upgradation technology selected for this project is AtmosPower's two stage patented system wherein the second stage serves the purpose of minimizing the methane losses and to achieve the recovery rate of more than 98%. It runs at very low pressure (0.7 barg) and hence offers the advantage of extremely low power consumption of around 0.22 kW per m³ of Raw Biogas. Its advantages are summarized in below table 1.2

Table 1.1 Advantages of two stage MPSA over other technologies

Conventional / Single Stage System	Two Stage MPSA system
Methane losses 10 – 30%	Methane losses < 2%
Methane purity in the product gas 90 – 95%	Methane purity in the product gas > 96%
0.3 – 0.4 kW/m ³ power consumption	0.20kW/m ³ power consumption
Requires careful precleaning (H ₂ S, H ₂ O removal) and post treatment sometimes	Comes with inbuilt precleaning units, can withstand higher impurities, and does not require and post treatment
High Opex and Replacement cost	Lower Opex and Replacement cost



Figure 1.3 Image of Atmos Power's Biogas Upgradation System

Process Flow Diagram:

